

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_cb1d52fa-fed\agent-a15b9dbc7dc5d5154.jsonl`  
- SHA-256: `4418d324ce65a58d65ef863c335f3997ae630ab07b49bf4694d6050cecccce810`  
- Source modified: `2026-06-14T08:26:14+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_cb1d52fa-fed`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T08:24:06.516Z)

THE PIVOT under study = "PERSONALIZATION-FIRST": instead of one universal model for all users, ship a BATTERIES-INCLUDED baseline that CONTINUALLY SELF-OPTIMIZES to a SINGLE user's own footage. As the user's corpus grows (10-15 videos), the on-device model + PER-USER CUE SELECTION (driven by the EXISTING LOVO / A-B experiment harness run on the user's OWN videos, with the user's CORRECTION LOOP as the per-user label source) becomes heavily tuned for THAT user ? deep analysis of THEIR rallies + shot types (serves, smashes, drives, drop shots). Per-user decisions like "the motion/person cue gives no F1 win for THIS user -> default it OFF for them." The model is optimized for the user's videos, NOT others (wins may hold anecdotally on other footage but weaker). Periodic "BATTERIES UPGRADES" improve the shipped baseline so it needs merely <5 videos to reach superior quality. Two user archetypes: (a) an economically-generous hobbyist who records sessions some days; (b) a professional with 100+ hours of their own video. CONTRAST = "generalization-first" (build one model that works on everyone's videos). The owner believes the harness + eval + A/B were DESIGNED to enable this roadmap.

DIMENSION: DATA / LABELS / PRIVACY / LICENSING. Read the guardrails in CLAUDE.md + docs/COMMERCIALIZATION.md if present. Analyze: the user's OWN videos as training data are commercial-CLEAN (no third-party copyright; paid-LLMs never a training source); on-device personalization = strong PRIVACY (data never leaves the machine; GDPR-friendly; the minors-exclusion + per-user opt-in guardrails still apply); the CORRECTION LOOP (user confirms/fixes detections) as the per-user LABEL source (pseudo-label then correct) ? elegant + already in v1 scope; and how ROORA'S ROLE SHIFTS ? from labeling a generic golden set to SEEDING THE BASELINE ("batteries") corpus, an oracle for the shipped baseline, not the per-user model. Return pros/cons + how it shapes the data strategy.

2. assistant (2026-06-14T08:24:10.576Z)

I'll analyze this pivot from the DATA / LABELS / PRIVACY / LICENSING dimension. Let me first read the relevant guardrails and strategy docs.

3. user (2026-06-14T08:24:14.048Z)

Already up to date.

---LS DOCS---

ANALYZERS_AND_RECONCILERS.md

BACKLOG.md

CODE_MAP.md

COMMERCIALIZATION.md

DEFERRED_HARDENING_PLAN.md

EVALUATION.md

EXECUTION_POLICY.md

EXPERIMENT_HARNESS.md

FIELD_VISIT_ANSWER_SHEET.md

FIELD_VISIT_PLAYBOOK.md

GOLDEN_SET_IMPLEMENTATION.md

GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PER_VIDEO_OUTPUT_LAYOUT.md
PLATFORM_ARCHITECTURE.md
PMF_FIELD_SIGNALS.md
PMF_MARKET_RESEARCH.md
PRODUCT_FIELD_ASSOCIATE_FLYER.md
QUALITY_ITERATIONS.md
README.md
ROADMAP.md
ROORA_LABELING_GUIDELINES.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_LOCALITY_MODEL.md
VIDEO_RECORDING_GUIDELINES.md
archives
mockups

4. user (2026-06-14T08:24:14.054Z)

```
1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30     counsel . ? future
31     Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33     order.)
34
35     | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
36     |---|---|---|---|---|---|
37     | | **? PENDING (needs action / not started)** | | | |
```

35 | C5 | ****Output codec royalties**** | ****P0**** | ? | Highlights are still encoded ****H.264 + AAC**** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure remains. | Switch output to ****AV1 + Opus**** (royalty-free) on an ****Ada-class GPU (L4/L40)**** for HW encode. See §4. Decision: provision Ada-class GPU? |

36 | C9 | ****Packaging / distribution**** | ****P0**** | ? | No `pyproject.toml`; the diagnostic `setup.py` shadows the build system. | Design in `DEFERRED\_HARDENING\_PLAN.md` (#7) ? ****awaiting your approval**** before code. |

37 | C12 | ****SaaS foundations (async/auth/tenancy)**** | ****P1**** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM\_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED\_HARDENING\_PLAN.md` (#16). |

38 | | ****? IN-FLIGHT (in progress / watch)**** | | | |

39 | C4 | ****Runtime AI strategy (A/B/C + owned model)**** | ****P0**** | ? | Live = ****C Gemini 2.5 Flash, paid tier****. Long-term = ****own a self-hosted generative video-QA model**** (fine-tuned open VLM) as primary, ****paid Gemini always-ready as fallback**** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model ****after**** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |

40 | C13 | ****Beachhead validation**** | ****P0**** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends ****India badminton academies**** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |

41 | C8 | ****Golden-set vendoring spend**** | ****P1**** | ? | ADR-006: ****Roora (Bengaluru)**** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |

42 | C3 | ****Shuttle-tracking weights provenance**** | ****P1**** | ? ? | WASB-SBDT ****code is MIT (clear)****; distributed ****checkpoints are academic-data-trained**** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial). | ****Ship-gating**** blocker (multiplies per sport). Resolve via an own-trained detector ****or**** a clean MIT/Apache detector swap ? NOTE: ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |

43 | C14 | ****Owned-model base licensing + training-data policy**** | ****P1**** | ? ? | Owned video-QA model must build on a ****commercial-clean (MIT/Apache-2.0/BSD), video-capable**** base ? primary ****Qwen3-VL (Apache-2.0)****, fallback ****InternVL3 (MIT)****. ****Gemma 4 = AMBER / bench-only**** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build target yet). ****Reject Gemma 3**** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). ****Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (competing-model clause). The moat is the ****consented dataset + eval harness****, not the model. | ****MIT/Apache-2.0/BSD**** for code+data+weights; ****open-model teachers only****; separate explicit ****training opt-in****; ****exclude minors****; per-clip consent ledger. See `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (authoritative). |

44 | C6 | ****Inbound HEVC decode**** | ****P2**** | ? ? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is ****cheap**** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48 | C2 | ****`supervision` ByteTrack pin**** | ****P2**** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | ****? COMPLETED**** | | | |

50 | C1 | ****AGPL / copyleft removal**** | ? | ? | ADR-005 shipped: AGPL `ultralalytics` YOLOv8 replaced with ****torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)****. No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

```

55
56 Verdicts updated against the current stack (`requirements.txt` + code). Original
57 2026-05-30 verdicts and the 25-claim verification transcript are in
58 `archives/MONETIZATION_AUDIT.md`.
59
60 | Dependency | License | SaaS obligation? | Current verdict | Notes |
61 |---|---|---|---|---|
62 | `~`ultralytics` + `yolov8n.pt`~` | AGPL-3.0 | Yes ($13) | ? **REMOVED** | Replaced per
ADR-005 ? no longer in the stack. |
63 | **torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen
replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`). |
64 | **supervision** (ByteTrack) | MIT | No | ? LIVE | Pinned `<0.30.0` ? migrate before
bump (C2). |
65 | **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open
(C3). |
66 | **MonoTrack / YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
67 | **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH |
Must bill the paid tier to keep user video private (C4). |
68 | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? |
Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |
69 | ffmpeg/ffprobe, google-genai, torch, opencv, numpy,
fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as
subprocess / imported; all permissive. |
70 | **obsreport** (reports) | permissive (private) | No | ? soft dep | Pin the
`git+ssh@vX` tag once the repo is published (BACKLOG). |
71
72 **Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)
73 are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
74 and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
75
76 ---
77
78 ## 3. Runtime AI strategy ? A/B/C
79
80 The semantic rally-boundary step is the one external/metered dependency. Three paths
81 (from the audit), with current standing:
82
83 - ** (A) Eliminate ? offline learned segmenter. ** ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? **target**
87 - ** (B) Self-host VLM. ** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - ** (C) Keep Gemini 2.5 Flash (paid). ** Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.

```

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111
112     **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113     output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114     *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116     ---
117
118     ## 5. Hosting unit-economics (C7)
119
120     Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122     - **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123       tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124       economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125     - **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126       VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127     - **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128       + light GPU. **Best steady-state margin.**
129
130     > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131     > **benchmark on the target VM before pricing the product.**
132
133     ---
134
135     ## 6. Product-market fit & the quality/data phase
136
137     ### Beachhead (from market research, 2026-06-07 ? see
138     [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139     The static-camera constraint is a **B2B filter**: the buyer must already record from a
140     fixed angle, which points away from the median casual player and toward coaches/clubs.
141     Ranked beachheads:
142     1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143     players,
144       formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
145       who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
146       not
147       the individual; needs INR-tuned pricing **and a Kannada/Hindi product** ? see
148       [`I18N_PLAN.md`](I18N_PLAN.md) (Bangalore-first localization, by design).
149     2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8%
150     YoY)
151       but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
152       head-to-head, not greenfield.
153     3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China
154     needs
155       local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
156
157     Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B
158     academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/
159     Spiideo; comparable to SwingVision software-only).
160
161     ### PMF risks (from the readiness review)
162     - **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
163       footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
164       narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
165     - **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;
166       per-rally tags, stats, and a correction loop are what make it sticky (C10).
167     - **Golden labels are the recurring blocker** across the offline segmenter, audio
168       fusion, and measurable quality ? the highest-leverage investment (C8).
169
170     ### Refined priority order (quality/data phase ? "focus on quality once new videos
171     land")
172     Carried from the readiness review; commercialization items above are tracked in
173     parallel.
174
175     1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos):

```

```

169         ingest the new batch, expand calibration/CV/regression baselines, make label
170         contribution + safe retraining easy and non-contaminating (eval/training separation).
171     2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract
172         shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics
land.
173     3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/
174         min-duration + `rally_gate`; validate WASB on real video; trial audio fusion
(ADR-007).
175     4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ?
176         `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
177     5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ?
178         validators/report rules; "your video is good enough" UX.
179     6. **Multi-sport maturation + recording best practices.**
180     7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
181     8. **Polish & distribution** (C9): installer story, first-run path, codec/output,
economics.
182
183     **Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
184     (flywheel + quality iteration on real data), then unification (#2) and deeper value
(#4).
185
186     ---
187
188     ## 7. Open questions needing owner / counsel sign-off ??
189
190     1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
191         insufficient (fallback). Quote-only.
192     2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
193         confirm each (C3).
194     3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
195         figures with counsel before relying on the "small SaaS under threshold" reading
(C5/C6).
196     4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
197         throughput (C7).
198     5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
199
200     ---
201
202     ## Changelog
203
204     - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
205         ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
206         "beachhead validation" workstream. Wired the summary into §6.
207     - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
208         audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
209         removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
210         the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
211         `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
212         exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
213
214     ---
215
216     ## See also
217     - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
beachhead; cited).
218     - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
verification transcript).
219     - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
220     - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
process).
221     - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
222     - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
223     - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
224     - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.

```

5. assistant (2026-06-14T08:24:16.883Z)

Now let me read the key docs on the experiment harness, correction loop, golden set vendoring (Roora's role), and owned-model plan to ground this analysis.

6. user (2026-06-14T08:24:17.853Z)

```

1      # Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by
design)
2
3      **Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` +
`tests/test_experiment.py`, shipped
4      2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest),
`compare_unlabeled()`
5      (SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.**
Phases below are
6      sequenced but each is a separate PR.
7
8      > **Companion docs:** the cues this experiments *on* are in
9      > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); the scoring mechanics are
10     > [EVALUATION.md](EVALUATION.md); the no-regression gate + golden labels +
`baseline.json` come from
11     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md); the modular-cue architecture is
**ADR-007**
12     > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the distilled
classifier is
13     > **ADR-004**. The 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
14
15     ---
16
17     ## 1. Context ? why this, why now
18
19     PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues,
all **default
20     OFF**. The owner's directive for *how* we evolve quality from here:
21
22     > "Evolve the codebase so we can keep finding new ways to improve quality and
experiment with them ?
23     > or add them as a signal ? without suffering any regressions if the idea doesn't work
out. Rolling
24     > back the code would be too dangerous."
25
26     So **experimentation and deployment must be decoupled.** The contract:
27
28     1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production
behavior.
29     2. Test it **offline against golden labels** (most experiments never touch the served
path at all).
30     3. Promote a variant to default **only on measured lift**, through a CI eval gate.
31     4. **"Rollback" = flip a config flag, never `git revert`**. The proven path is a pinned
variant that is
32     always present.
33
34     **The pleasant surprise (Explore audit, 2026-06-13):** ~80% of this harness already
exists and is
35     simply not composed *as a system*. The missing piece is a thin **variant/experiment
layer + the
36     discipline**, not a platform. §4 is the exact code surface so the next session does not
re-discover it.
37
38     ---
39

```

```

40     ## 2. The thesis ? separate expensive DETECTION from cheap DECISION
41
42     The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per
43     frame" (shuttle
44     WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python
45     part is the
46     **decision**: "given those signals, where are the rallies" (windowing + gates + the
47     classifier).
48
49     > **Run detection once per video, persist all per-cue signals + per-candidate features,
50     then re-score
51     > any number of variants OFFLINE ? deterministically, with no GPU and zero production
52     risk.**
53
54     This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do
55     exactly this on
56     stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence
57     substrate.
58     ? Most experiments are pure re-scoring of stored data; they never enter the served
59     pipeline.
60
61     ---
62
63     ## 3. What already exists (the audit ? do NOT rebuild)
64
65     Exact public surface, with `file:line`. **This table is the anti-duplicate-work
66     artifact** ? start here.
67
68     ### 3.1 Scoring (variant-agnostic ? grades any `List[Interval]`)
69     `backend/eval/metrics.py`
70     - `interval_iou(a, b) -> float` (:15) ? temporal IoU of two `(start,end)` intervals.
71     - `match_intervals(preds, gts, iou_threshold=0.5) -> MatchResult` (:48) ? greedy 1-to-1
72     IoU matching.
73     - `score(preds, gts, iou_threshold=0.5, match_result=None) -> Score` (:104) ? P/R/F1,
74     mean IoU, boundary errors; `Score.as_dict()`.
75     - `pr_curve(preds, gts, thresholds=None) -> List[Score]` (:138).
76
77     ### 3.2 A/B + cross-validation primitives (the comparison engine already exists)
78     `backend/eval/calibration.py`
79     - `improvement_report(predict_fn, baseline_config, tuned_config, gts, iou_threshold=0.5)
80     -> dict` (:148) ? **before/after metrics + %?. This IS the A/B delta.**
81     - `leave_one_video_out(videos, configs, metric="f1", iou_threshold=0.5) -> dict` (:113)
82     ? **LOVO: pick on other videos, score on held-out video (honest cross-video).**
83     - `cross_validate(predict_fn, configs, gts, k=5, metric="recall", iou_threshold=0.5) ->
84     dict` (:72) ? within-video k-fold overfit guard.
85     - `select_best(predict_fn, configs, gts, metric="f1", iou_threshold=0.5) -> (Config,
86     float)` (:61).
87     - `evaluate(preds, gts, iou_threshold=0.5) -> Score` (:41); `kfold_indices(n, k)` (:46).
88     - LOVO video dict shape: `{"name": str, "predict_fn": PredictFn, "gts": [Interval]}`.
89
90     ### 3.3 No-regression gate + baselines (the promotion gate already exists)
91     `backend/eval/regression.py`
92     - `regress(db, video_id, ground_truth, stage="final", iou_threshold=0.5, detector=None,
93     tolerance=DEFAULT_TOLERANCE, baseline_path=DEFAULT_BASELINE_PATH) -> RegressionResult` (:104) ?
94     score stored preds vs GT, flag if P **or** R drops > tolerance.
95     - `regress_with_provider(...)` (:160) ? same, fetching GT via the annotation provider.
96     - `save_baseline(video_id, stage, precision, recall, f1, ..., gt_fingerprint=None)`
97     (:80) . `load_baselines(path) -> dict` (:68).
98     - `RegressionResult` (:43): `regressed`, `reasons`, `gt_hash`, baseline P/R,
99     `tolerance`, `has_baseline`; `as_dict()`.
100     - Default baseline file: `eval_baselines/regression_baseline.json` (:33).
101
102     `run_regression.py` ? CI-gatable CLI, `main(argv)` (:52). **Exit codes: 0 ok / 1
103     regression / 2 no-DB / 3 no-baseline.** Flags: `--video-id --tier --annotations
104     --stage{candidates,final} --detector --iou --tolerance --baseline --update-baseline --allow-

```

```

missing-baseline --json`.
84
85     ### 3.4 Pinnable model artifact
86     `backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2):
`fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75),
`predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON)
(:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
87
88     ### 3.5 Offline re-scoring substrate (persist once, re-score forever)
89     `backend/infrastructure/database.py`
90     - `add_candidate_segment(video_id, detector, start_time, end_time, chunk_index=None,
confidence=None, stats=None, detection_params=None) -> Optional[int]` (:240) ?
`stats`/`detection_params` stored as JSON; `candidate_segments` table cols incl. `stats`,
`detection_params`, `human_label`, `confirmed_segment_id` (schema :97).
91     - `query_candidate_segments(video_id, detector=None, only_unlabeled=False) ->
List[dict]` (:277) ? returns rows with deserialized `stats`.
92
93     ### 3.6 Label assignment + feature glue
94     `backend/eval/training.py`
95     - `label_candidates(candidate_intervals, gt_intervals, iou_threshold=0.5) -> List[int]`
(:50) ? y-labels by IoU match to golden (same matcher as the evaluator).
96     - `build_training_set(candidate_rows, gt_intervals, iou_threshold=0.5,
feature_fn=extract_yolo_features) -> (X, y, intervals)` (:64) ? pluggable `feature_fn`.
97
98     ### 3.7 Existing LOVO drivers to mirror (not duplicate)
99     - `backend/eval/distill_local.py` ? `run(specs, iou=0.5, threshold=0.5, model_out=None)`
(:98), CLI `python -m backend.eval.distill_local --manifest videos.json --model-out ...`;
`FEATURE_NAMES = [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]` (:40);
already does LOVO over Gemini silver labels.
100     - `backend/eval/calibrate_local.py` ? `run(specs, iou=0.5, metric="f1")` (:77);
`LocalConfig = (velocity_thresh, merge_gap, min_window_duration, min_crossings)`; grid + LOVO.
101     - `backend/eval/calibrate_wasb.py` ? `run(...)` , `load_golden_gts()`;
`backend/eval/gt_loader.py::load_gt_intervals(csv_path, *, strict=True)` is THE canonical golden
reader.
102
103     ### 3.8 Registries + config knobs (variant selection)
104     - `backend/pipeline/segmenters/base.py` ? `SegmenterRegistry` (:35), singleton
`segmenter_registry` (:63); registered: `motion`, `gemini`, `yolo_hybrid`, `tracknet_hybrid`,
`wasb_hybrid`.
105     - `backend/annotations/base.py` ? `AnnotationProviderRegistry`, singleton
`annotation_registry`; providers `file`, `auto`.
106     - `backend/config/models.py::IndexingConfig` ? `default_segmenter` (:65, default
`"yolo_hybrid"`), `detector_model` (:72), `detector_score_threshold` (:73),
`yolo_velocity_threshold` (:74), `yolo_frame_skip` (:75), `min_segment_duration` (:68),
`dead_time_merge_gap` (:69), `motion_threshold` (:67). **No `min_crossings` knob yet** (see
fusion P2).
107
108     ### 3.9 Confirmed ABSENT (no duplication risk)
109     Explore found **no** existing experiment / variant / shadow / A-B *machinery* ? only a
stray "A/B test"
110     aspiration comment in `backend/pipeline/detectors/base.py` and an "experiment E1" note
in
111     `AUDIO_RALLY_DETECTION_PLAN.md`. The `regress()` + baseline path is the canonical
comparison mechanism.
112
113     ---
114
115     ## 4. The variant abstraction (the one thin thing to add)
116
117     A **`Variant` = a declarative recipe**, not a code branch:
118
119     ```
120     Variant = (segmenter_name, IndexingConfig overrides, enabled cues + per-cue params,
classifier model path/hash)
121     ```

```

```

122
123     JSON-serializable; selected via the existing `segmenter_registry` +
`IndexingConfig.default_segmenter`.
124     **Control** = a *pinned, frozen* variant (recipe + model hash) that is always registered
and is the
125     default. Adding a new cue = a new Variant differing by one flag ? the control recipe is
untouched.
126
127     ---
128
129     ## 5. Three experiment modes
130
131     ### 5.1 Offline A/B (primary, label-driven, zero production risk)
132     `compare(videos, variant_A, variant_B)` ? a thin driver composing **existing**
functions:
133     `query_candidate_segments()` (load persisted features) ? re-score each variant ?
`calibration.improvement_report()`
134     + `calibration.leave_one_video_out()` + `metrics.score()`, graded vs golden
(`gt_loader.load_gt_intervals`).
135     Reports per-video **and** LOVO-honest aggregate IoU/P/R/F1 deltas. **This is the A/B
test, and it needs
136     almost no new scoring code ** ? only the glue.
137
138     **Honest reporting (default ON, additive):** `compare(..., honest_stats=True)` also
attaches a "honest"
139     block ? per-metric bootstrap **CIs**, a paired **?F1 CI + exact sign test** (C1), and
**F1@k +
140     segment-count ratio ** (C4) ? *alongside* the bare-mean fields (existing output
unchanged; `honest_stats=False`
141     restores the legacy shape). A bare LOVO mean at n?6 is not promotable on its own; the
lift is real when the
142     ?F1 CI clears 0 **and** the sign test agrees. Wiring of the course corrections in
143     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13/C1+C4 (helpers:
`backend/eval/eval_stats.py`,
144     `segmentation_metrics.py`); `record_experiment` carries the honest ?F1 into the
leaderboard.
145
146     ### 5.2 Shadow (for cues that touch detection, e.g. the person detector)
147     The new cue runs and **persists its signal** into the candidate
`stats`/`detection_params`, but the
148     *served* decision stays control. Offline you then ask "what would variant B have
scored?" via §5.1 ?
149     B never affects output until it wins. This is how a detection-touching change (not just
a re-scoring
150     change) earns its way in without risk.
151
152     ### 5.3 Promotion gate (no-regression, CI-enforced)
153     `regression.regress()` + `baseline.json` + `run_regression.py` exit codes. A variant
becomes the new
154     default **only if** it doesn't regress the golden set beyond `tolerance` (exit 1
blocks). Control is
155     pinned ? **rollback = flip `default_segmenter`/variant config, never a code revert.**
156
157     ---
158
159     ## 6. No-regression guarantees (the heart of the owner's ask)
160
161     - **New cues/signals default OFF** ? per-cue enable flag in `IndexingConfig`; merged
code can't change
162     behavior until explicitly enabled.
163     - **Control variant pinned** ? frozen recipe + classifier model hash, always registered
& default.
164     - **Promotion only via the eval gate** ? `run_regression.py` in CI (exit 1 blocks); no
silent default change.
165     - **Experiments are records** ? variant-spec hash ? per-video + LOVO scores + label-set

```

```

hash + timestamp,
166     persisted to a small JSON **experiment leaderboard** (generalizes `baseline.json`).
Answers "which
167     signals ever helped, on which video slices," and makes every result reproducible.
168     - **Decoupled events** ? "merge an experiment" and "change the default" are separate,
independently
169     revertible actions.
170
171     ---
172
173     ## 7. What to build later (gap list ? thin, additive, owner-gated)
174
175     1. ~**`backend/eval/experiment.py`** ? a `Variant` dataclass + `compare(videos, a, b)`
composing the $3
176     functions (no new scoring math).~ **DONE (2026-06-13).** Shipped `Variant` (+ pinned
`CONTROL`),
177     `compare` (labeled A/B with LOVO-honest learned-variant training, mirroring
`distill_local`),
178     `compare_unlabeled` (the **SxS-on-new-videos** mode ? agreed / A-only / B-only via
`match_intervals`,
179     no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), and
`load_video_candidates`
180     (the `query_candidate_segments` bridge). JSON variant recipes first as recommended; a
`VariantRegistry`
181     only if the count grows. 19 tests incl. the no-regression invariant.
182     2. **Persist per-cue features** into candidate `stats` (person: `players_in_court`,
`two_sided_motion`,
183     `in_court_ratio`, `shuttle_player_colocation`; shuttle-state: `net_crossings`,
`direction_reversals`;
184     audio: hit times) so any future variant re-scores offline ? ties to
`MULTI_SIGNAL_FUSION_PLAN.md` $3.2/$4.
185     3. **Per-cue enable flags** in `IndexingConfig` (default `False`); `default_segmenter`
already exists.
186     Add the `min_crossings` knob (fusion P2) here too.
187     4. **CI promotion lane** using `run_regression.py` against the golden `baseline.json`;
write the
188     experiment-leaderboard JSON.
189
190     ---
191
192     ## 8. Phasing (owner-gated; ONE PR per slice, off `master`)
193
194     - **P0 ? this doc**, indexed in `docs/README.md`, cross-linked from
`MULTI_SIGNAL_FUSION_PLAN.md` and
195     `GOLDEN_SET_VENDORING.md`. *(this deliverable; docs-only, no code.)*
196     - **P1 ? `experiment.py` `compare()`** ? **DONE (2026-06-13):** thin wrapper over
`calibration` +
197     `metrics` + `query_candidate_segments`; pure-logic, fully tested in this dev container
(`compare`,
198     `compare_unlabeled`, `record_experiment`, the no-regression invariant).
`compare_unlabeled` adds the
199     SxS read on **new unlabeled footage** the owner asked for (run control + candidate on
a fresh video,
200     diff where they agree/diverge ? two reels via the existing `VideoCompiler` is the GPU-
side follow-on).
201     - **P2 ? per-cue enable flags + feature persistence** into candidate `stats` (default
OFF) ? couples
202     with fusion P1/P2.
203     - **P3 ? CI promotion gate**: `run_regression.py` lane on the golden `baseline.json`;
experiment-leaderboard JSON.
204
205     ---
206
207     ## 9. Scope guardrail
208

```

```

209     **Single-tenant / local-first** (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment
registry + the
210     existing harness ? explicitly **NOT** MLflow / W&B / a service. Keep it the size of
`baseline.json`.
211
212     ---
213
214     ## 10. Verification
215
216     - **Offline, runnable in this dev container (no GPU/torch needed for the glue):** unit
test that
217         `compare()` is deterministic on synthetic candidate rows; a **no-regression invariant
test** asserting
218         "all new-cue flags OFF ? predictions byte-identical to control"; a CI-gate test
asserting
219         `run_regression.py` exit codes (0 ok / 1 regress / 3 no-baseline).
220     - **Machine-side (owner, GPU/WSL):** persist candidates once on a real clip, run
`compare(control,
221     fusion)` LOVO, confirm the reported lift matches a manual `metrics.score()` spot-
check.
222     - **Docs (P0):** links resolve; indexed in `docs/README.md`; cross-linked from
223         `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`.
224     </content>
225

```

7. user (2026-06-14T08:24:18.189Z)

```

1     # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3     **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
greenlights a specific
4     milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a
training source).
5     **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6     (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8     ## Design principle (owner directive): *envision now, build later*
9     Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10    now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11    event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12    punting its *seam* is not.
13
14    ## North Star (reframed per PR #61 review)
15    - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16    recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17    - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18    contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19    - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20    roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21    - **The moat = the consented dataset + the eval harness, not the weights.**
22
23    ## Hard guardrails (non-negotiable ? "never corrupted")
24    - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25    (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-

```

restrictive: Gemma 3 (viral),

26 ****Gemma 3n**** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.

27 - ****Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.**** Also forbidden: public grounding-CoT datasets that

28 were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision (a real, binding cost).

29 - ****Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.****

30 - ? ****Base-model pretraining provenance is unauditible for *every* open VLM (incl. Qwen3-VL).**** Our "no paid-LLM

31 training" rule is enforceable at ***our fine-tuning data*** layer, not the base's pretraining ? an ****explicit owner**

32 **risk-acceptance**** is required to use any open VLM. (Governed by the base's license, not our pipeline.)

33

34 **## The model strategy (verified 2026-06-09)**

35 ****Framework ? base**** (the Gemma-4-vs-ms-swift clarification): a ***framework*** is the forge (runs SFT/GRPO, emits

36 weights); a ***base*** is the metal you forge. You pick one framework and feed it any clean base.

37 - ****Framework: ms-swift (Apache-2.0)**** ? the one forge that does native ****video + audio + timestamp-grounding +**

38 GRPO/RFT with a custom video reward** in one stack. Keep ****unsloth**** (Apache-2.0 core; *avoid its AGPL Studio UI*)

39 as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-time snapshot ? re-verify at impl.

40 - ****Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)**** ? long-video context, purpose-built

41 ****timestamp emission**** (second-level localization), multi-turn video chat. Re-confirm the exact size's LICENSE

42 pre-release. ****Gemma-4 stays AMBER and bench-only**** ? its Apache grant is likely but its Prohibited-Use posture is

43 unconfirmed (counsel needed), ***and*** its video/timestamp capability is disputed; not the primary extractor.

44 ****InternVL3 (MIT/Apache)**** = clean fallback if a Qwen blocker appears. ? base swap is cheap ***only within**

45 timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-specific).

46 - ****START WITH THE TAL HEAD, NOT THE VLM**** (the decisive call): fine-tuned VLMs ****miss strict temporal boundaries****

47 (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").

48 ****Boundary accuracy IS the product.**** The head (``rally_seq_proto.py``) is already de-risked on our corpus (held-out

49 AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super (\$0), is ~1M params (trivially compressible later),

50 and ****doubles as a pseudo-label teacher + the honest baseline the VLM must beat.**** So: ****head now ? VLM later****,

51 strictly sequenced. When the VLM comes, prefer ****two-stage**** (VLM coarse-proposes, head refines boundaries) over

52 wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*

53

54 **## Conversational video-QA ? build the bookkeeping NOW, punt the model**

55 ****Punt the conversational *feature***** until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI

56 over a weak extractor exposes the weakness). ****Reject end-to-end-VLM-answers-each-turn**** (expensive; bad at exact

57 counting; the strong teachers are license-forbidden). ****But build the *seam* now**** (the brief requires it):

58 - ****Architecture = "extract once, query many":**** EXTRACTION (TAL head ? later VLM) emits structured timestamped

59 spans ? a durable ****per-video EVENT STORE**** ? a ****structured QUERY layer**** (text-to-SQL + ffmpeg clip-cut). The

```

60      cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a
61      VLM job** (SQL counts exactly where VLMs hallucinate).
62      - **Designed-now seam:** make the **per-video event-index schema a first-class data
contract** alongside the M0.1
63      corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count,
ending_reason/fault_tags,
64      derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy
(e.g. `service_fault`) up front**
65      even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration):
`play_segments` today has
66      `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights`
isn't a table yet ? so it's an
67      *extension with new columns + a highlights table*, not a no-op. **Punt** the NL
parser/agent until quality clears the bar.
68      - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our
structured index (allowed ?
69      it reasons over our data, it is not trained on its outputs).
70
71      ## Multisport (single sport-agnostic model)
72      - **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned
across sports (sport = an input
73      feature / per-sport calibration). The collector + indexer are already sport-generic. ?
**ACTION: rename the repo off
74      "badminton" ASAP** (owner emphatic) ? consistent with this.
75      - **What multisport changes = DATA + DETECTORS (scale up, the accepted "good
problem"):** each new sport needs its own
76      labeled matches (split by match) + a **clean trajectory detector** to feed the head's
features. **The binding blocker
77      is per-sport detectors:** WASB (badminton) has the unresolved **C3** checkpoint-
provenance issue, and **clean MIT/
78      Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet
identified.** Sequence by public-data
79      richness: **badminton ? tennis/TT ? pickleball/padel.**
80      - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81      and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82      concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport
83      corpus exists. Don't cite RacketVision as evidence.
84
85      ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
shell
86      - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
offline/batch, no latency hot path.
87      - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-
process (ffmpeg, WASB-in-WSL, cloud
88      GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
single-digit-ms wins dwarfed by
89      ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
harness + provider registry.
90      - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91      Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92      runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93      target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94      - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95      separate service. **None exists today.**
96
97      ---

```

```

98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
102     *greenlight item*
103     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
104     schema (the conversational seam)
105     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
106     ending_reason/fault_tags,
107     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
108     MATCH), trajectory_ref,
109     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL
110     ActivityNet-JSON · VLM clip?QA),
111     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
112     (system-of-record; extends PR #13).
113     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
114     enforced; **no Gemini-sourced rows in any training view**.
115
116     ### M0.2 ? Grow the corpus · **DONE (n=6), running** · *owner action + my scoring loop*
117     6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-
118     match + LOVO recorded; the
119     "contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club
120     match ? margin. **Bottleneck =
121     data + calibration, not method.** Keep growing per new sport.
122
123     ### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) .
124     *greenlight item*
125     - **(i) PURGE the Gemini-silver TRAINING path** ? the live no-paid-LLM-training
126     violation is
127     **`backend/eval/distill_local.py`** (it trains a classifier on Gemini-silver CSV
128     labels ? `output/local_rally_model.json`)
129     and the threshold distillation in `calibrate_local.py`. (NB:
130     `backend/eval/training.py::label_candidates` is a *pure,
131     source-agnostic* function ? there is no store by that name; the earlier
132     "training.label_candidates" pointer was wrong.)
133     Nothing Gemini-trained is in git (artifacts live under gitignored `output/`), but the
134     path that regenerates them is one
135     command away ? retarget to human + open-teacher (own-model/WASB) labels; Gemini =
136     eval/reference/fallback only.
137     - **(ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint
138     provenance is a **commercial-ship
139     blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a
140     clean MIT/Apache detector swap.
141     ? **own-weights is an un-scoped NEW workstream, not ROADMAP Phase 4:** per ADR-002 our
142     temporal labels can't train the
143     shuttle detector (it needs per-frame coordinate labels we don't have). **Multiplies
144     per sport** (incl. any TT-on-WASB use).
145     Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
146
147     ### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) · *greenlight item*
148     Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
149     **ActionFormer (MIT, 1:1
150     rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not*
151     the Commons-Clause repo),
152     reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item):
153     a FLOOR of "beat the honest n=6
154     heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU**
155     target for "very high P/R" (e.g.
156     F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory.
157     Trains on the GPU/WSL box.
158
159     ---
160
161     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
162     Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it

```

becomes the rally/highlight

138 extractor (local, MIT). If not, the gap (more data/features) is the next signal ? no overfitting to declare victory.

139

140 ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus + explicit go)

141 - **Gen-1:** concatenate fully-local cue channels onto the head. ? **Audio is NOT a promising cue here** ? our E1 probe

142 (PR #35) found it weak (~2.2× discrimination, AUC ~0.6) and the foreground-audio idea also failed this session; keep it

143 parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet first).

144 - **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a temporal-IoU-vs-golden reward,

145 our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on Modal (no-access) / RunPod-Secure.

146 - **Hard pre-serving gates (unmeasured, load-bearing):** the **\$/call crossover** (self-hosted VLM vs Gemini) **a real

147 VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs ? **budget is the

148 largest open risk and is currently unquantified**). Don't stand up an owned serving stack until these clear.

149

150 ---

151

152 ## Decisions / status

153 | # | item | status |

154 |---|---|---|

155 | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4 harness** | awaiting owner |

156 | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |

157 | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ? |

158 | 4 | Collector = corpus + event-store system-of-record | **owner: confirm** ? |

159 | 5 | Rename repo off "badminton" | **owner action** ? |

160 | 6 | Conversational-QA: build seam now, punt model | **rec accepted** ? |

161

162 ## Open risks / unknowns (none block Phase-0 research)

163 Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R" target undefined · per-sport clean

164 trajectory detectors not yet identified · base-model pretraining provenance unauditale (risk-acceptance) · real VLM

165 VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-train (minors-excluded) for cloud

166 training · on-device 4B latency on 8 GB unmeasured.

167

168 ---

169 *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing discipline holds: the committed

170 next platform slice remains the async job model (single-tenant); only the M0.1?M0.4 foundation is safe to start.*

171

8. assistant (2026-06-14T08:24:21.797Z)

Now let me read the golden-set vendoring doc (Roora's role) and the correction-loop / annotation provider surface, plus check guardrails on consent/minors/opt-in.

9. user (2026-06-14T08:24:22.288Z)

1 # Golden-Set Generation & Vendor Process (v1)

2

3 **Status:** Design v1 · **Date:** 2026-05-30 · **Vendor shortlist added 2026-05-30 (\$6)**

4 **Owner decisions captured:** licensed/owned video only · rally `[start,end)` depth ·

tiered turnaround (fast automated + slow human) · vendor bake-off · **Bangalore/India-local vendor preference**

5

6 This document defines how we produce **golden-set ground truth** (hand-verified rally boundaries) for the badminton highlights product, how we **select annotation vendors**, and how that same machinery powers **on-demand / surprise regression testing** of the production tool.

7

8 > **Companion docs:** the scoring mechanics live in EVALUATION.md; the annotated-corpus rationale is [ADR-002](archives/decisions/DECISIONS.md); the decision record for **this** process is **ADR-006**. Video-capture quality requirements (a dependency of trustworthy golden sets) are tracked separately in VIDEO_RECORDING_GUIDELINES.md. The ``regress()`` + ``baseline.json`` machinery in §5 is generalized into a **variant A/B + promotion gate** in EXPERIMENT_HARNESS.md.

9

10 > **Update (2026-06-02):** §6 shortlist done; the founder selected **Roora** (Bengaluru) for the **initial paid pilot** ? a **single-vendor pilot first**, not the full bake-off below. Roora's site/capability/location are verified; **quality** is TBD via the pilot. **Step 0** (the annotation guideline) is authored ? collector repo ``docs/annotation/`` (``ROORA_BRIEF.md`` rally CSV schema + ``ANNOTATION_GUIDE.md`` illustrated guide + intake-manifest & NDA templates). Pilot scope is **3 sports** (badminton, pickleball, table tennis); the collector's ``import-local`` ingestion for all three is built. Still pending: the internal **gold key** and the vendor-dependent **GS-3 `HumanVendorProvider`** (GS-1/GS-2/GS-4 already landed ? see GOLDEN_SET_IMPLEMENTATION.md).

11

12 ---

13

14 ## 1. The keystone insight: one scorer, three jobs

15

16 A vendor's golden-set output is a list of rally ``[start,end)`` intervals ? **structurally identical** to a prediction from our pipeline. So the **exact same** deterministic scorer in ``backend/eval/metrics.py`` (``interval_iou`` ? ``match_intervals`` ? ``score``) grades all three of:

17

18 1. **Product vs. ground truth** ? the precision/recall/F1/IoU we already report (``EVALUATION.md``).

19 2. **Vendor vs. our internal answer key** ? the quality gate during vendor selection.

20 3. **Annotator vs. annotator** ? inter-annotator agreement (IAA), our proxy for "is this even a well-defined task?"

21

22 This symmetry is what makes the whole system small, auditable, and extensible. **No new scoring code is needed** ? only a new **source** of intervals (the vendor).

23

24 ---

25

26 ## 2. Why this also solves the privacy nuance

27

28 The monetization goal (hosted API) carries a "someone views the user's video" concern. This process sidesteps it **by construction**:

29

30 - **Golden sets** are built only from footage we license or own ? starting with the founder's **constantly-growing personal collection** of match videos, plus ShuttleSet (ADR-003) and any purpose-recorded clips. **No end-user uploads ever flow to a vendor.**

31 - Regression tests therefore run against a **representative held-out corpus**, not literal user content.

32 - All video egress to any third party passes through a **single provider boundary** (§4), so the security controls (DPA, no-reuse, delete-on-delivery, access logging) live in exactly one place.

33

34 > **Honest tradeoff:** a licensed/owned corpus may not perfectly mirror real users' cameras, courts, and lighting. A green regression on our corpus is therefore **strong evidence**, not a **guarantee**, for production traffic. Revisit if field behavior diverges from corpus behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by steering users toward capture conditions our corpus represents.

35

36 ---

```

37
38     ## 3. Selecting vendors ? the bake-off
39
40     ### Step 0 ? Write the annotation guideline (highest-leverage artifact)
41     Before contacting anyone, write a precise, example-rich definition of what a rally
42     is:
43     - Start trigger: e.g. first serve motion vs. shuttle leaving the racket ? pick one,
44       show clips.
45     - End trigger: shuttle hits floor/net/out, or umpire call.
46     - Edge cases: lets/replays, service faults, between-point walking, coaching pauses,
47       mid-rally net-cord, broadcast replays and graphics.
48
49     Rally boundaries are genuinely subjective; ambiguity here is the #1 cause of bad
50     golden sets. This guideline belongs conceptually with the per-sport `SportProfile`, so it
51     generalizes to tennis/cricket later.
52
53     > Done (2026-06-02): authored as the Roora brief + illustrated guide in the
54     collector repo `docs/annotation/`. Defines the rally CSV schema and the three golden rules ?
55     ignore all dead time; include every rally (incl. service faults & lets); a shot = one
56     racket/paddle/bat contact ? plus an `ending_reason` taxonomy (winner / forced_error /
57     unforced_error / service_fault / let / other, shared across net-separated sports) and per-sport
58     detail incl. singles & doubles for badminton & pickleball.
59
60     ### Step 1 ? Build a small internal gold key
61     One trusted expert (initially: the founder) hand-annotates 5?10 licensed clips to
62     near-perfect. This is the answer key vendors are graded against ? kept private during the bake-
63     off.
64
65     ### Step 2 ? Run the bake-off (parallel pilot)
66     Send the same pilot clips to 2?3 shortlisted vendors (see §6, favoring
67     Bangalore/India), withholding the key. Grade every submission with our own harness:
68
69     | Dimension | How we measure it | Tool |
70     |---|---|---|
71     | Accuracy | vendor CSV vs. gold key ? P/R/F1, mIoU, boundary error | `eval.score()`
72     |
73     | Consistency | 2 annotators per vendor ? pairwise IoU (IAA) | `interval_iou` |
74     | Cost | quoted ? or $ per video-minute | vendor quote |
75     | Turnaround | actual vs. promised, per tier | observed |
76     | Ops friction | file-drop vs. API; compliance with our CSV format |
77     | Security | DPA signed; DPDP/ISO posture; delete-on-delivery | contract review |
78
79     ### Step 3 ? Decide on measured quality-per-dollar
80     The bake-off numbers are the decision ? not sales decks. Low IAA at a vendor means
81     either a weak vendor or a vague guideline (Step 0); investigate before blaming the vendor.
82
83     ### Step 4 ? Onboard the winner(s)
84     Sign DPA + the security controls in §4; codify the guideline as their spec; agree SLAs
85     for both tiers (§5).
86
87     ---
88
89     ## 4. The extensible architecture ? an `AnnotationProvider` registry
90
91     This codebase already uses pluggable registries (validators, segmenters, AI providers,
92     sports). The idiomatic move is the same shape for annotation sourcing:
93
94     ```python
95     class AnnotationProvider(ABC):
96         def submit(self, video_ref, guideline, sport) -> str:          # returns job_id
97             ...
98         def poll(self, job_id) -> str:                                  #
99         queued|running|done|failed
100
101     ...

```

```

83         def fetch(self, job_id) -> List[Interval]:                # canonical
[(start,end), ...]
84         ...
85         ...
86
87         `fetch()` returns the same `List[Interval]` that `annotations.load_annotations()`
already produces, so everything downstream is unchanged.
88
89         Three implementations cover today plus both tiers:
90
91         | Provider | Role | Tier |
92         |---|---|---|
93         | `FileProvider` | Reads a pre-made CSV (formalizes today's manual / ShuttleSet flow) |
? |
94         | `HumanVendorProvider` | The bake-off winner; authoritative ground truth, corpus
growth, deep audits | Slow (days) |
95         | `AutomatedProvider` | An independent strong model labels rallies programmatically |
Fast (hours) |
96
97         The tiers are just two providers behind one interface, selectable per call.
98
99         ---
100
101         ## 5. The regression automation (the explicit ask)
102
103         > "given a video, get a golden set created and verify its precision and recall against
the production tool"
104
105         This becomes a thin wrapper over machinery that already exists:
106
107         ```python
108         def regress(video, tier="fast", tolerance=0.05):
109             gt                = annotation_registry.get(tier).fetch(                # NEW: the provider
110                 annotation_registry.get(tier).submit(video, GUIDELINE, sport))
111             predictions = process(video)                # existing production
pipeline ? DB
112             s                = eval.score(predictions, gt)                # existing metrics.py,
unchanged
113             regressed      = (s.recall < baseline[video].recall - tolerance or
114                 s.precision < baseline[video].precision - tolerance)
115             if regressed: alert(video, s, baseline[video])
116             return s
117         ```
118
119         Only one new component (the provider). `run_eval.py`, `harness.py`, and `metrics.py`
are reused verbatim. Snapshot a `baseline.json` per corpus video so "regression" is a measured
delta, not a vibe.
120
121         Triggers (all feed the same `regress()`): CI on pipeline changes · on-demand (the
"surprise" case) · scheduled cron · pre-release gate.
122
123         ### ?? The oracle problem (a real correctness trap)
124         The fast-tier `AutomatedProvider` must be a different model lineage than production**.
If production becomes the offline ActionFormer segmenter (ADR-004) and the oracle is also
ActionFormer, they share blind spots ? the test shows green while both are wrong. Pairings that
stay honest:
125
126         - production = offline learned segmenter ? fast-tier oracle = paid Gemini full-video
(different lineage).
127         - production = Gemini hybrid ? fast-tier oracle = a different VLM or the slow human
tier.
128
129         Treat the fast tier as a smoke alarm, not ground truth: any fast-tier "regression"
escalates to the slow human tier for confirmation before it blocks a release.
130

```

131 ---
132
133 ## 6. Vendor shortlist (Bangalore/India-favoring)
134
135 > **Researched 2026-05-30 via a lean manual pass** (targeted web search + direct site/source verification ? NOT the heavy automated workflow, which failed). **Confidence is labelled per row.** **No pricing is listed because none could be verified** ? every vendor in this space quotes per-project, so ?/clip is an offline-conversation item (see §6.3). Treat single-source rows as *leads to verify on the call*, not vetted recommendations.
136
137 ### 6.1 Tier-1 candidates ? verified India presence + video capability (talk to these first)
138
139 | Vendor | Location (verified) | Video / temporal labeling | Notes for the call |
Confidence |
140 |---|---|---|---|
141 | **iMerit** | **Bengaluru** office (A101 & A118, 43 Residency Rd, 560025; tech hub opened Oct 2022); also Kolkata HQ + US | Yes ? image/video/text/audio; explicit Sports & Biomechanics domain on their site | Largest & most enterprise-ready (5,500+ annotators); has a real **Bangalore** footprint for in-person coordination; ask for a sports/temporal-segmentation reference + a paid pilot. Higher-touch ? likely higher cost. | **High** |
142 | **TELUS International AI** (ex-**Playment**) | Playment was **Bangalore-founded** (2015); acquired by TELUS 2021, team retained | Yes ? built on Playment's 2D/3D image+video+LiDAR CV pipeline | Strong CV/video pedigree, Bangalore roots; but now a large global BPO ? confirm they'll take a **small startup pilot** and who the India delivery contact is. | **High** (origin) / **Med** (current India ops shape) |
143 | **Taskmonk** | **Bengaluru** | Yes ? "Video" listed as a labeling modality; platform **++** managed-service workforce | Mid-size, India-local, does both tooling and service ? potentially the best **cost/agility** fit for a startup pilot. Verify temporal/event labeling specifically (their heritage is e-commerce catalog tagging). | **Med-High** |
144
145 ### 6.2 Tier-2 ? India video-annotation services, location/specialts to confirm on contact
146
147 | Vendor | Stated location | Note | Confidence |
148 |---|---|---|---|
149 | **Infolks** | India (Kerala-based per third parties) | Targets startups/SMBs; image/video/text/audio; positions on cost-efficiency ? good cheap-pilot candidate | **Med** |
150 | **Analytics** | India + USA (since 2018) | General data-annotation incl. video; widely listed | **Med** |
151 | **Objectways** | HQ Scottsdale AZ; **India delivery in Coimbatore** (not Bangalore) | Does video/LiDAR; India delivery but not Bangalore-local | **Med** |
152 | **Roora** (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragr.com`) | Image/video/text annotation; sports industry listed. **Selected for the initial paid pilot (2026-06-02)** ? existence/capability/location verified; **quality TBD via the pilot**. | **Verified** (capability) · pilot pending |
153 | **Tika Data** / **Zuru** | Claimed **Bangalore** (single-source listicle each) | Smaller/startup annotation shops ? unverified beyond one directory mention; verify they exist & do temporal video before investing time | **Low** |
154
155 ### 6.3 Explicitly NOT verified / out of scope
156 - **Pricing for everyone** ? all quote-only; do not assume. Get per-video-minute (or per-rally) quotes during the offline conversations.
157 - **Karya (Bengaluru)** ? confirmed local and reputable, but **speech/text/translation only, no video** ? **excluded** for our use case.
158 - **Sports/temporal track record** ? beyond iMerit's stated "Sports & Biomechanics" domain, vendors' specific *rally-boundary / temporal-event* experience is unconfirmed marketing copy; probe with the pilot (§3).
159 - **DPDP / DPA / ISO posture** ? not verified per-vendor; make it a checklist item in the bake-off (§3 Step 2 security row).
160
161 ### 6.4 Recommendation
162 For the **offline conversations**, start with **iMerit** (Bangalore-local, enterprise QA, sports domain) and **Taskmonk** (Bangalore-local, likely more startup-friendly cost/agility) as the two primary contacts, with **TELUS/Playment** as a third if you want a heavyweight CV

pedigree. Run the \$3 bake-off across whichever 2?3 agree to a paid pilot, and decide on measured quality-per-rupee. **Verify every "Confidence: Med/Low" claim on the call before committing.**

163

164 > **Update (2026-06-02):** the founder opted to start with **Roora** (Bangalore-local, startup-friendly) as a **single-vendor pilot** rather than the iMerit/Taskmonk multi-vendor bake-off above. iMerit / Taskmonk / TELUS remain the fallback shortlist if the Roora pilot doesn't clear the \$3/\$4 quality bar.

165

166 > Sources (manual pass, 2026-05-30): iMerit Bengaluru hub ? imerit.net press / PRNewswire (Oct 2022) + contact listings; Playment/TELUS ? BusinessWire & Inc42 (2021 acquisition, Bangalore-founded); Taskmonk & Karya locations ? company sites + AnalyticsIndiaMag / IndiaAI; others ? AnalyticsIndiaMag / Ziloservices / vendor sites. Pricing intentionally omitted (unverifiable).

167

168 ---

169

170 ## 7. Suggested phasing

171

Phase	Deliverable	Depends on
0	Annotation guideline + internal gold key (5?10 clips)	nothing ? can start now
1	Vendor bake-off ? pick winner(s)	\$6 research + guideline
2	`AnnotationProvider` registry + `FileProvider` + `HumanVendorProvider`	winner onboarded
3	`AutomatedProvider` (independent oracle) + `regress()` + baseline snapshots	Phase 2
4	Wire surprise triggers (CI / on-demand / scheduled)	Phase 3

179 Phase 0 is unblocked today and is the highest-leverage step regardless of vendor choice.

181

182 > **Status (2026-06-02):** Phase 0 guideline ? authored (collector `docs/annotation/`); internal gold key still pending. Phase 1 is running as a **single-vendor Roora pilot** across **3 sports** (badminton, pickleball, table tennis) rather than a multi-vendor bake-off. Phase 2's `FileProvider` (GS-1) and Phase 3's `regress()` + automated oracle (GS-2/GS-4) have already landed; the main remaining slice is the **vendor-dependent `HumanVendorProvider`** (GS-3), with trigger-wiring (GS-5) after it.

183

184 ---

185

186 ## 8. Open questions

187 1. **Guideline authorship** ? who owns the canonical rally definition, and is one expert's key enough, or do we want 2-expert consensus for the answer key itself?

188 2. **Corpus representativeness** ? how well does the owned/licensed collection mirror eventual user capture? (Ties directly to VIDEO_RECORDING_GUIDELINES.md.)

189 3. **Baseline drift** ? when production legitimately improves, how/when do we re-snapshot `baseline.json` so old baselines don't raise false regressions?

190 4. **Fast-tier oracle cost** ? per-video Gemini cost at regression frequency (see `archives/MONETIZATION\_AUDIT.md` \$4 for the per-video estimate).

191

10. user (2026-06-14T08:24:22.922Z)

1 # Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

2

3 **Status:** **P1 DONE** (2026-06-13) ? the person cue is extracted to

4 `backend/pipeline/detectors/person\_detector.py` (`PersonDetector`), consumed by `yolo\_hybrid` with no

5 behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a separate PR.

6 **Owner decisions captured** in [\$9](#9-open-decisions-for-the-owner).

7

8 > **Companion docs:** the 2-layer mental model is HOW_RALLY_DETECTION_WORKS.md;

9 > the modular-fusion architecture this builds on is **ADR-007** (archives/decisions/DECISIONS.md);

```

10     > the permissive-dependency rule is **ADR-002 / ADR-005**; the audio cue (the #1 next
cue, already
11     > scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUD
IO_RALLY_DETECTION_PLAN.md).
12     > The vendor labels that gate the learned path come from
[GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)
13     > and the labeling spec in [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md).
14     > **How each cue here is added/tested without production regression** ? A/B, shadow
eval, and the
15     > promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)
(every P1?P4 cue
16     > below lands flag-gated, default OFF, promoted only on measured LOVO lift).
17
18     ---
19
20     ## 1. Context ? why this, why now
21
22     Rally detection today runs **one cue at a time**. Each segmenter independently turns a
single signal
23     into rally windows:
24
25     | Segmenter | Signal | File |
26     |---|---|---|
27     | `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory)
| `backend/pipeline/detectors/tracknet_runner.py` |
28     | `yolo_hybrid` (the current `default_segmenter`) | person velocity (torchvision +
ByteTrack) | `backend/pipeline/segmenters/yolo_hybrid.py` |
29     | `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |
30     | `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |
31
32     **There is no fusion.** The owner's directive: make the detector **use the shuttle *and*
the humans
33     together** ? and be ready to fold in more signals (audio, later pose) ? delivered
34     **MIT / Apache-2.0 / BSD-clean** so we can ramp commercially without a licensing
rethink.
35
36     This is **not greenfield** ? it operationalizes work the repo already blessed:
37
38     - **`HOW_RALLY_DETECTION_WORKS.md` open question:** **Fusion: combine shuttle-motion
(WASB) with
39     player-motion for static cams.**
40     - **ADR-007** already designed a **modular "rally layer"** that fuses cues **outside* the
frozen WASB
41     model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks
up the parked
42     person cue and generalizes the fusion seam so future cues are "register a producer,"
not "re-touch
43     the segmenter."
44     - **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate,
explicitly
45     labelled in its own docstring as **Gate A in the over-fire fix spectrum (**B = player-
stance state
46     machine, future**). The person cue in this plan **is that named-but-unbuilt Gate
B.**
47
48     ### The owner's "held-shuttle" false positive ? what's actually going on
49
50     The owner observed a "rally" that was just a player **holding / flicking the shuttle in
hand** between
51     points. This is the **exact* failure `rally_gate.py` was written to kill:
52
53     > **WASB over-detects: when a player flicks/tosses the shuttle back to the opponent for
service after
54     > losing a point, that single trajectory looks like a rally. The discriminator is
structural: a real

```

```

55     > rally has the shuttle crossing the net MULTIPLE times; a single service feed crosses
~once."*
56
57     **The structural fix already exists and is validated offline ? but it is NOT wired into
the production
58     runtime path.** `rally_gate.apply_gate(..., min_crossings=2)` is called only from the
**eval drivers**
59     (`distill_local.py`, `eval_partial_labels.py`, `export_wasb_segments.py`,
`calibrate_local.py`); there
60     is **no `min_crossings` knob in `IndexingConfig`** and the live segmenters don't apply
it. So the
61     single cheapest, highest-confidence win in this whole plan is **P2's first step: fold
the existing
62     Gate A into production.** Everything else (the person cue / Gate B, the learned fusion)
is robustness
63     on top of that.
64
65     ### Why humans help ? and the honest risk
66
67     The beachhead videos are **static tripod** (Bangalore academies,
[PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md)),
68     so player presence/motion is a usable cue. ADR-001's warning was specifically about
**broadcast pan**
69     cameras, where player-motion fooled the old heuristic ? *that failure mode does not
apply to a fixed
70     court cam.* But we keep **the shuttle as the camera-invariant primary** so the general
bet stays robust
71     on other camera types. Humans then add:
72
73     - **Precision** ? reject shuttle-on-floor / warm-up knock-ups / held-shuttle feeds when
the court
74     isn't occupied by ?2 players in a play posture.
75     - **Recall** (via the learned classifier) ? features that help bridge shuttle
occlusion/blur gaps.
76
77     ---
78
79     ## 2. Core architecture ? one rally-fusion layer, cues are swappable producers
80
81     Formalize ADR-007's diagram. Every cue is a **black box that emits a time-aligned
signal** (per-frame
82     activity and/or per-window features); **fusion happens in *our* layer, never inside a
frozen model.**
83
84     ```
85     frames ?? WASB (frozen, MIT)      ?? shuttle trajectory ???
86     frames ?? person detector (ours)  ?? tracks + boxes ??????
87     audio  ?? hit detector (ours)     ?? hit events ???????????? RALLY LAYER (ours) ??
rallies
88     [frames ?? pose/stance (ours; parked #3) ?? stance ???????? feature + decision fusion
89     ```
90
91     - **Shuttle = primary, camera-invariant.** Windows still **seed** from
92     `TrackNetRunner.trajectory_to_action_windows()` (`tracknet_runner.py:234`). Other cues
*enrich and
93     adjudicate* shuttle-seeded windows; they do **not** replace the shuttle seed. Person
runs only over
94     **shuttle-seeded candidate windows + padding**, not the whole video ? which bounds the
added cost.
95     - **A small `RallyCue` contract.** Each producer yields `(frame_idx ? features)` + per-
window stats.
96     Adding a cue = register a producer; the existing `SegmenterRegistry`
(`segmenters/base.py:35`) is the
97     template for the pattern. Keep the single-cue segmenters registered as fallbacks.
98

```

```

99      ---
100
101      ## 3. The two fusion modes (both ? per the owner decision)
102
103      ### 3.1 Decision-level gate (safe baseline, ships first, no labels needed)
104
105      ```
106      rally_active = shuttle_motion ? (?? persons in court zone ? recent person track)
107      ```
108
109      The `? recent-track` clause tolerates 1?2 frame person-detector misses so the AND-gate
110      can't cause a
111      recall cliff. This composes with the existing net-crossing Gate A:
112      ```
113      keep window ? net_crossings ? 2                (Gate A ? EXISTS in rally_gate.py, eval-
114      only today)
115      ? (?? in-court players ? recent track)    (Gate B ? the person cue, NEW)
116      ```
117      Gate A kills the service-feed / held-shuttle case structurally; Gate B adds court-
118      occupancy evidence
119      for the cases a single trajectory can still fool (e.g. a knock-up that does cross the
120      net). No training
121      data required ? this is the label-free fusion that ships before any golden labels exist.
122
123      ### 3.2 Feature-level (primary, the learned path)
124
125      The cues emit a small set of aggregated, scale/translation-invariant per-window
126      features ? NOT raw
127      coordinates. Raw pixel coordinates would memorize *this* court/camera and fail to
128      generalize off a
129      still-small labeled set; counts / ratios / zone-membership / two-sidedness transfer.
130
131      What already feeds the classifier (`distill_local.py:40`, `FEATURE_NAMES`):
132      `duration, peak_velocity, mean_velocity, active_ratio, net_crossings` ? note
133      net_crossings is
134      already a feature** (computed via `rally_gate.gate_windows`).
135
136      Person features to add (~4, deliberately few):
137
138      | Feature | Meaning | Rally signal |
139      |---|---|---|
140      | `players_in_court` | median count of persistent in-court tracks | ? 2 (singles) / 4
141      (doubles); 0?1 = dead time |
142      | `two_sided_motion` | are moving players on both net-halves? | rally = two-sided;
143      one person feeding = one-sided |
144      | `mean_player_motion` | aggregate normalized player velocity | rally = sustained;
145      resting/standing = low |
146      | `in_court_ratio` | fraction of frames with ?2 in-court players | track stability vs
147      flicker |
148      | `shuttle_player_colocation` | fraction of frames the shuttle sits inside/adjacent to a
149      person box (hand region) | held = high; in-flight = mostly free space ? the direct held-
150      shuttle discriminator |
151
152      These join the shuttle features in the candidate `stats` JSON
153      (`database.py::add_candidate_segment`) and the feature vector of the existing
154      distilled classifier ?
155      the tiny numpy-only L2 logistic regression in `backend/eval/classifier.py` (ADR-004),
156      NOT a new net.
157
158      It scores each candidate window `P(real rally)` and replaces the Gemini call at runtime.
159
160      What the model learns: the weights on those few features ? e.g. down-weight a
161      shuttle-motion
162      window with 0?1 in-court players / one-sided motion / high shuttle-colocation (warm-up,

```

```

floor shuttle,
148     held-shuttle feed); up-weight ?2 players, two-sided, sustained motion, shuttle in free
flight.
149
150     **Small-N discipline (non-negotiable):**
151     - Keep to **~4 person features** ? more features on a still-small labeled set overfit;
that's exactly
152     why the model stays a logistic regression, not a net.
153     - Use the **leave-one-video-out (LOVO)** protocol that `distill_local.py` **already
implements**: train
154     on the other videos' candidates, predict the held-out *video*, score vs its labels ?
never a
155     held-out *window* from the same video (windows in one video are correlated; that
flatters the score).
156     - Training labels today are **Gemini silver labels** (distilled offline, ADR-004/007 ?
Gemini is a
157     teacher, never a runtime dependency, never a training-data *source* for owned
weights); **golden
158     labels are the honest measuring stick**, and as the Roora golden corpus grows they can
also become
159     training labels. Until the labeled set is larger, treat the learned weights as
160     **calibration / direction-check, not "a fully general model"*** ? the decision-level
gate (§3.1) is
161     the shippable, label-free fusion in the meantime.
162
163     > *(Gated on golden labels existing ? the same blocker ADR-007 flags for audio. See
164     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md).)*
165
166     ---
167
168     ## 4. Shuttle-state: in-flight vs in-hand (the misclassification, mapped to real code)
169
170     The "shuttle moved" seed conflates a shuttle **in flight** with a shuttle **carried in
hand**. The
171     discriminators, and where each lives:
172
173     | Signal | In-flight | Held / static | Status |
174     |---|---|---|---|
175     | `net_crossings` | ? 1 (usually several) | ~0 (held) / ~1 (single feed) | **EXISTS**
(`rally_gate.py`); eval-only ? wire to prod |
176     | `shuttle_player_colocation` | low (free space) | high (near a hand) | **NEW** (needs
the person cue) |
177     | `direction_reversals` | several (racket contacts) | ~0 | derivable from trajectory;
partial overlap with crossings |
178     | `peak shuttle speed` | high | walking speed | derivable from existing velocity |
179
180     So the in-flight-vs-held fix is **mostly Gate A (already built) + the new colocation
feature from the
181     person cue**. **No new label column is needed** ? the golden labels already start a
rally at the
182     *serve-contact frame* and treat all pre-serve / held-shuttle time as dead time (see
183     [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md)), so held-shuttle moments
are negatives
184     by construction and are the ground truth that proves the detector wrong.
185
186     ---
187
188     ## 5. Adversarial mitigations (integrating safely)
189
190     - **Court-region mask** ? polygon around the court; drop persons outside it (spectators,
coaches,
191     adjacent courts). Config-supplied per camera; ~1 ms. Source: a manual per-camera
polygon first;
192     the **4 court corners + net-line** that Roora labels once per video (see the labeling
doc) build

```

```

193     this mask **and** the net split that `two_sided_motion` / `net_crossings` need.
194 - **Player-count constraint** ? expect **2 (singles) / 4 (doubles)** persistent in-court
tracks;
195     suppress transient/extra detections. The `singles_or_doubles` manifest flag sets the
expectation.
196 - **Temporal association** ? reuse `supervision` **ByteTrack** (already in
`yolo_hybrid`); reject
197     singleton detections; smooth across frames.
198 - **Multi-court rejection** ? proximity to the primary court region (homography
optional, later).
199 - **Domain shift (COCO frontal ? side/overhead court)** ? start **box-only** (robust);
keep the
200     detector swappable. Fine-tuning a person detector is **parked** ? we have no in-house
person labels,
201     and the **paid-LLM/training guardrail + minors-exclusion** apply to any future person-
training data.
202 - **Fusion-degradation guard** ? shuttle stays primary; person is gate + feature, never
the sole
203     trigger; the single-cue segmenters remain registered fallbacks.
204
205 ---
206
207 ## 6. Box vs pose
208
209 **Box-first.** Occupancy + count + two-sided motion + shuttle-colocation is enough for
both the gate
210     and the features, and box detection is the cheap, robust win. **2D pose is parked behind
the same
211     `RallyCue` interface ? pose (MediaPipe Pose / RTMPose, both Apache-2.0) buys serve-
ready stance
212     gating and shot cues later, at the cost of a second model. This matches ADR-007 parking
pose as the
213     #2/#3 cue.
214
215 ---
216
217 ## 7. Performance
218
219 The person detector is the **existing in-process torchvision model** run at **frame-skip
375**
220     (`yolo_frame_skip` default 3) with track interpolation between sampled frames. The
shuttle keeps
221     running in its separate WSL process ? the two are **separate processes**, so a "shared
backbone" isn't
222     free; budget realistically as the existing `yolo_hybrid` cost minus skip savings. Person
runs **only
223     over shuttle-seeded windows + `ai_handoff_padding`**, not the whole video, which bounds
the cost.
224
225 ---
226
227 ## 8. Licensing (the guardrail table ? all green)
228
229 Every dependency stays **MIT / Apache-2.0 / BSD** (ADR-002 / ADR-005; `ultralitics` was
dropped for
230     AGPL-3.0).
231
232 | Cue / model | Code | Weights / data | Verdict |
233 |---|---|---|---|
234 | Shuttle: WASB-SBDT, TrackNetV3 | MIT | frozen badminton weights | ? (ADR-001/002) |
235 | Person (default): torchvision Faster R-CNN / RetinaNet / FCOS | BSD/MIT | COCO-
pretrained | ? already in repo (`_DETECTOR_FACTORIES`) |
236 | Person (escalation): **YOLOX**, **RT-DETR / RT-DETRV2** | Apache-2.0 | COCO /
Objects365 | ? |
237 | Pose (parked): **MediaPipe Pose**, **RTMPose / MMPose** | Apache-2.0 | COCO-structure

```

```

| ? |
238 | Tracking: `supervision` ByteTrack | MIT | n/a | ? already in repo |
239 | **Banned** | **Ultralytics YOLOv5/8/11 (AGPL-3.0)** . **MonoTrack (Adobe, non-comm)**
. **YOLO-NAS (Deci, non-comm)** | ? | ? |
240
241 **COCO caveat to record:** COCO *annotations* are **CC BY 4.0** (commercial OK *with
attribution*);
242 COCO *images* are Flickr-ToS (per-image). Pretrained-**weight** use is clean; record the
attribution.
243
244 ---
245
246 ## 9. Open decisions (for the owner)
247
248 1. **Court-region mask source** ? manual per-camera polygon (simple, ship first) vs auto
court
249 homography/line detection (later). **Recommend manual first**, fed by Roora's court-
corner labels.
250 2. **Person detector default** ? keep torchvision (zero new deps, BSD/MIT) vs adopt
YOLOX/RT-DETR
251 (Apache, faster) now. **Recommend torchvision first**, escalate only if eval shows a
speed/accuracy gap.
252 3. **Pose now or parked** ? **recommend parked** behind the interface (box-first).
253 4. **Golden labels** ? feature-level fusion (P3) is **blocked on golden rally labels**;
the source is
254 the VLC `rally-annotator` + the Roora vendor pilot
([GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)).
255
256 ---
257
258 ## 10. Phased execution (owner-gated; ONE PR per slice, off `master`)
259
260 - **P0 ? this design doc** +
[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md), indexed in
261 `docs/README.md`, cross-linked from `HOW_RALLY_DETECTION_WORKS.md` and ADR-007. *(this
deliverable;
262 docs-only, no code.)*
263 - **P1 ? extract the shared person cue:** ? **DONE (2026-06-13).** Lifted the
torchvision detector +
264 ByteTrack + velocity out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
265 (`PersonDetector`); `yolo_hybrid` now orchestrates only and delegates to `self.person`
(no behavior
266 change ? regression-safe, proven by the unchanged process_video output tests). New
267 `tests/test_person_detector.py`; suite +1 green; person_detector is type-clean (left
mypy quarantine
268 for the moved code). The fusion segmenter (P2) reuses this same producer over shuttle-
seeded windows.
269 - **P2 ? the FusionSegmenter (Gate A + Gate B + person-feature persistence):** ? **DONE
(2026-06-14).**
270 Shipped `backend/pipeline/segmenters/fusion_hybrid.py:FusionSegmenter` (registry
`fusion_hybrid`,
271 **default-OFF** ? `default_segmenter` unchanged), subclassing the trajectory engine
via two identity
272 hooks (`_enrich_candidate_stats`, `_select_for_handoff` ? wasb/tracknet stay byte-
identical, adversarially
273 verified). It seeds windows from the shuttle substrate, **always persists
`net_crossings`** (Gate A signal),
274 and ? when `indexing.fusion.cue_flags.person` is on ? runs the person cue over each
window's frames
275 (`PersonDetector.detections_per_frame`) and persists the ~5 person features
(`fusion_features.py`,
276 pure + unit-tested). Served gating is optional + default-OFF: `fusion.min_crossings`
(Gate A) /
277 `fusion.min_players` (Gate B) skip sub-threshold windows from the handoff while

```

```

**persisting the full
278     superset** (offline substrate intact). Court mask = optional `fusion.court_polygon`
(None ? count-only).
279     Lift to be measured via the harness; held-shuttle-drop + person-feature values are
GPU/owner-side.
280     - **P3 ? feature-level (learned) fusion:** add the person features (incl.
`shuttle_player_colocation`)
281     to `distill_local` / `classifier`; recalibrate IoU on the golden set; report **LOVO
lift on held-out
282     videos**. *(Gated on golden labels.)*
283     - **P4 ? additional cues:** wire **audio** (ADR-007's #1, already scaffolded in
284     `backend/pipeline/audio/hit_detector.py`) and later **pose**, through the same
`RallyCue` interface.
285     Each is add-a-producer, not a refactor.
286
287     ---
288
289     ## 11. Verification
290
291     - **Offline, runnable in this dev container (no GPU/torch/cv2):** unit tests for the
fusion gate logic
292     and `RallyCue` aggregation (court-mask filter, player-count constraint, `? recent-
track` tolerance)
293     and the shuttle-state signals (net-crossing count ? extends the existing `rally_gate`
tests ? and
294     person-colocation) with **synthetic trajectories + person boxes**. A key-license
assertion test
295     (no `ultralytics` import; detector factories stay torchvision/Apache).
296     - **Machine-side (owner, GPU/WSL ? the full pipeline can't run in this container):** run
the
297     `FusionSegmenter` on a real static-tripod clip; confirm **held-shuttle / warm-up /
spectator false
298     positives drop** vs shuttle-only (this should reproduce the owner's observed bug and
show it fixed ?
299     start by confirming Gate-A-in-production alone fixes it); **LOVO IoU lift on the
golden set** for the
300     learned path.
301     - **Docs (P0):** links resolve; both docs indexed in `docs/README.md`; this doc cross-
linked from
302     `HOW_RALLY_DETECTION_WORKS.md` (the fusion open-question) and ADR-007.
303

```

11. assistant (2026-06-14T08:24:26.255Z)

I now have strong grounding. Let me check the correction-loop surface (annotations/human label) and the consent/minors guardrails in the schema/code to confirm what's actually built vs. designed.

12. user (2026-06-14T08:24:27.824Z)

```

Found 19 files
tests\test_database.py
backend\infrastructure\database.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
tests\test_eval_stats.py
tests\test_score_calibration.py
tests\test_segmentation_metrics.py
backend\eval\eval_stats.py
backend\tools\export.py
backend\pipeline\segmenters\fusion_hybrid.py
backend\pipeline\segmenters\yolo_hybrid.py
tools\cleanup_caches.py
tests\test_cleanup_caches.py
backend\eval\gt_loader.py

```

```
training\gen0\harness.py
tests\test_training_gen0.py
tests\test_eval_classifier.py
setup.py
backend\reporting\harvest.py
```

13. user (2026-06-14T08:24:28.384Z)

```
Found 12 files
docs\CODE_MAP.md
docs\STORAGE_SHARING_MODEL.md
CLAUDE.md
docs\VIDEO_LOCALITY_MODEL.md
docs\PLATFORM_ARCHITECTURE.md
docs\UI_ALPHA_EXECUTION_PLAN.md
backend\eval\calibrate_local.py
docs\UI_PROPOSAL.md
docs\HOSTING_PLAN.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
```

14. assistant (2026-06-14T08:24:32.017Z)

Let me check the correction-loop reality in the DB schema and how consent/opt-in/minors are handled, plus the platform/storage-sharing model where personalization is discussed.

15. user (2026-06-14T08:24:35.916Z)

```
backend\infrastructure\database.py:109:             confirmed_segment_id INTEGER,
backend\infrastructure\database.py:110:             human_label TEXT,
backend\infrastructure\database.py:114:             FOREIGN KEY
(confirmed_segment_id) REFERENCES play_segments(id) ON DELETE SET NULL
backend\infrastructure\database.py:279:             "UPDATE candidate_segments SET
confirmed_segment_id = ? WHERE id = ?",
backend\infrastructure\database.py:299:             query += " AND human_label IS NULL"
```

16. user (2026-06-14T08:24:36.153Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-2-
docs\STORAGE_SHARING_MODEL.md-3-Status: DRAFT ? owner + Claude co-design session pending;
nothing here is ratified.
docs\STORAGE_SHARING_MODEL.md:4-Date: 2026-06-12 (v2 adds: model-tier axis, social layer,
on-device end-state, abuse
docs\STORAGE_SHARING_MODEL.md-5-economics ? per owner's second brief) · Source: owner briefs
(voice note `Umiya Velociti.m4a`
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-6-pending
transcription) + UI_PROPOSAL · HOSTING_PLAN · PMF research · NEXT_STEPS quality table.
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\STORAGE_SHARING_MODEL.md-20-|---|---|---|---|
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-21-|
Storage (the meter) | 2 GB library | 10 GB+ / pooled | Mostly local disk; cloud = shareable
library only |
docs\STORAGE_SHARING_MODEL.md:22:| Model (rides the tier) | Owned model only ? never
Gemini | Best path (two-stage refine / frontier) | Best owned model, on-device |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-23-|
Compute (the cost) | Our GPU, capped runs | Our GPU, priority | Their GPU (USB-drive
workflow) |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-24-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\STORAGE_SHARING_MODEL.md-110-|---|---|---|---|
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-111-| Library
```

| 2 GB | 10 GB+ | Pooled across students |
docs\STORAGE_SHARING_MODEL.md:112:| Model | Owned only | Best path + priority queue | Best
owned, **on-device** |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-113-|
Processing | N matches/mo cap | Higher cap | Unlimited (their hardware) |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-114-| Social |
Full | Full | + coach overlay, athlete profiles |
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-131-locally
(this repo already *is* that pipeline) and syncs only reels + metadata to the cloud
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\STORAGE_SHARING_MODEL.md-132-library/profile. Cloud cost for the heaviest users ?
storage + social only.
docs\STORAGE_SHARING_MODEL.md:133:- **The licensing guardrail becomes the product moat:** Gemini
can never ship on-device; the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-134- owned
model (Qwen3-VL base, Apache-2.0 / MIT-only deps) CAN. Commercial-clean licensing was
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md-135- the
prerequisite all along.

17. assistant (2026-06-14T08:24:40.596Z)

I have everything I need. The correction loop is structurally present (`human\_label`,
`confirmed\_segment\_id` columns) but consent/opt-in/minors are design-only guardrails (no schema
columns yet ? only reserved in the M0.1 contract). The harness (`compare`, LOVO,
`record\_experiment`) and per-cue flags are real. Roora = licensed/owned-only corpus, no user
uploads. Let me synthesize the data/labels/privacy/licensing analysis.

18. user (2026-06-14T08:26:03.451Z)

Structured output provided successfully

19. assistant (2026-06-14T08:26:14.012Z)

I've completed the analysis of the PERSONALIZATION-FIRST pivot along the DATA / LABELS / PRIVACY
/ LICENSING dimension and returned it via StructuredOutput.

Key grounding from the codebase (all file paths absolute on read):

- The data substrate is real and shipped: correction-loop columns (`human\_label`,
`confirmed\_segment\_id`) in `backend/infrastructure/database.py:109-110`, the pseudo-label flow
via `query\_candidate\_segments`, and the full per-user A/B+LOVO harness
(`backend/eval/experiment.py`, `compare`/`leave\_one\_video\_out`/`record\_experiment` with
bootstrap-CI + paired sign test) per `docs/EXPERIMENT\_HARNESS.md` and recent commits (#124?127).
The owner's claim that the harness was designed to enable this is materially supported.
- The privacy/licensing guardrails (consent / opt-in / minors-exclusion) are design-only ?
reserved in the M0.1 contract (`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`) and stated in
CLAUDE.md, but no consent/opt-in/minor columns exist in the live schema yet.

The headline tensions I surfaced: (1) user footage as training data is the cleanest licensing
posture the project can have and dissolves the no-paid-LLM-training blocker for the per-user
path; (2) the privacy win is real only on-device, which is explicitly deferred ? near-term cloud
personalization re-opens DPA/Art.9; (3) the correction loop as a label source carries selection
bias and catastrophic small-N (n=10-15 is below the project's own promotability bar), making
per-user cue-ablation decisions statistically thin and risking disabling structural guards like
the held-shuttle Gate A; (4) Roora correctly shifts from per-user oracle to baseline-corpus
seeder/certifier ? but the baseline still needs an independent generic golden set, so vendoring
spend doesn't actually shrink.